

Introduction to Variational Inference

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1 Variational Inference

Variational Inference is a method for approximating a posterior distribution by a simpler tractable family of distributions. It uses optimization to select a distribution q that is closest to the true posterior in some metric. In practice, variational inference can be much faster than MCMC, though sometimes less accurate in representing posterior uncertainty.

The basic idea behind variational inference is very simple. Consider the standard Bayesian setup with data given by y and parameters given by θ . The prior is $f_\theta(\theta)$ and the likelihood is $f_{y|\theta}(y)$. The posterior is then:

$$f_{\theta|y}(\theta) = \frac{f_\theta(\theta)f_{y|\theta}(y)}{\int f_\theta(\theta)f_{y|\theta}(y)d\theta}.$$

The denominator is the marginal density $f_y(y)$ of y . The marginal density $f_y(y)$ is also known as the **Evidence**. Using the marginal density for the denominator, we get

$$f_{\theta|y}(\theta) = \frac{f_\theta(\theta)f_{y|\theta}(y)}{f_y(y)}.$$

The posterior is usually intractable because of the integration involved in the computation of the Evidence (denominator). In variational inference, one chooses a simpler family of distributions $\mathcal{Q}$ and then employs optimization techniques to choose a distribution in $\mathcal{Q}$ that is as close as possible to the posterior $f_{\theta|y}(\theta)$. The distance-measure used to measure closeness here is most commonly the following Kullback-Leibler divergence:

$$KL(q||f_{\theta|y}) = \int q(\theta) \log \frac{q(\theta)}{f_{\theta|y}(\theta)} d\theta.$$

Recall that the KL divergence between two densities q and p is given by $\int q \log(q/p)$. It is always nonnegative and it equals zero if and only if $q = p$. It is not symmetric in general (i.e., $KL(q||p) \neq KL(p||q)$).

The optimization that we need to solve therefore is:

$$\operatorname{argmin}_{q \in \mathcal{Q}} KL(q||f_{\theta|y}). \quad (1)$$

The choice of this specific KL divergence is mostly for technical convenience as it makes the resulting optimization tractable. Observe that

$$KL(q||f_{\theta|y}) = \int q(\theta) \log \frac{q(\theta)}{f_{\theta|y}(\theta)} d\theta = \int q(\theta) \log \frac{q(\theta)}{f_{y,\theta}(y, \theta)} d\theta + \log f_y(y). \quad (2)$$

The second term above (which involves the usually intractable $f_y(y)$) does not depend on q so the optimization (1) is equivalent to:

$$\operatorname{argmin}_{q \in \mathcal{Q}} \int q(\theta) \log \frac{q(\theta)}{f_{y,\theta}(y, \theta)} d\theta.$$

The above objective function is called the Variational Free Energy, or simply, Free Energy $F(q)$:

$$F(q) = \int q(\theta) \log \frac{q(\theta)}{f_{y,\theta}(y, \theta)} d\theta = - \int q(\theta) \log \frac{f_{y,\theta}(y, \theta)}{q(\theta)} d\theta. \quad (3)$$

So the main goal in Variational Inference is to minimize the Free Energy. The negative of the Free Energy is called the Evidence Lower Bound (ELBO):

$$\text{ELBO}(q) = -F(q) = \int q(\theta) \log \frac{f_{y,\theta}(y, \theta)}{q(\theta)} d\theta. \quad (4)$$

So the main goal in Variational Inference can also be said to be the maximization of the ELBO. The name ELBO comes from the fact that $\text{ELBO}(q)$ always satisfies:

$$\text{ELBO}(q) = \int q(\theta) \log \frac{f_{y,\theta}(y, \theta)}{q(\theta)} d\theta \leq \log f_y(y). \quad (5)$$

This is because of the expression (2) which says that the difference between $\log f_y(y)$ and $\text{ELBO}(q)$ equals the Kullback Leibler divergence $KL(q \| f_{\theta|y})$ which is nonnegative. Because $f_y(y)$ is called the evidence and $\text{ELBO}(q)$ gives a lower bound for the logarithm of the evidence, it is given the name Evidence Lower Bound. It is also easy to see that if we maximize $\text{ELBO}(q)$ over *all* probability densities q , then we obtain equality in (5):

$$\sup_q \text{ELBO}(q) = \log f_y(y). \quad (6)$$

The above follows from (2) because

$$\text{ELBO}(q) = \log f_y(y) - KL(q \| f_{\theta|y}).$$

Because $f_{y,\theta}(y, \theta) = f_{\theta}(\theta) f_{y|\theta}(y)$, the ELBO has the following alternative expression:

$$\begin{aligned} \text{ELBO}(q) &= -F(q) \\ &= \int q(\theta) \log \frac{f_{y,\theta}(y, \theta)}{q(\theta)} d\theta \\ &= \int q(\theta) \log \frac{f_{y|\theta}(y) f_{\theta}(\theta)}{q(\theta)} d\theta \\ &= \int q(\theta) \log f_{y|\theta}(y) d\theta - \int q(\theta) \log \frac{q(\theta)}{f_{\theta}(\theta)} d\theta \\ &= \int q(\theta) \log f_{y|\theta}(y) d\theta - KL(q \| f_{\theta}) \end{aligned} \quad (7)$$

In other words, $\text{ELBO}(q)$ represents the average likelihood (with respect to q) minus the KL divergence between q and the prior.

Usually, the main tasks in Variational Inference are (a) to choose a class of distributions $\mathcal{Q}$, and (b) to maximize $\text{ELBO}(q)$ (or, equivalently, to minimize $F(q)$) over $q \in \mathcal{Q}$. We will see how to do this via some examples.

2 Bayesian Logistic Regression

Consider again the logistic regression setting where we observe data $(x_i, y_i), i = 1, \dots, n$ with $x_i \in \mathbb{R}^d$ and $y_i \in \{0, 1\}$. The likelihood model is:

$$y_i \mid x_i, \beta \sim \text{Bernoulli} \left(\frac{e^{x_i^T \beta}}{1 + e^{x_i^T \beta}} \right)$$

and we take the improper uniform prior for β . We then have:

$$\begin{aligned} f_{y, \beta}(y, \beta) &= \prod_{i=1}^n \left(\frac{e^{x_i^T \beta}}{1 + e^{x_i^T \beta}} \right)^{y_i} \left(1 - \frac{e^{x_i^T \beta}}{1 + e^{x_i^T \beta}} \right)^{1-y_i} \\ &= \prod_{i=1}^n \frac{\exp(y_i x_i^T \beta)}{1 + \exp(x_i^T \beta)} = \exp(\ell(\beta)) \end{aligned}$$

where $\ell(\beta)$ is the log-likelihood is

$$\ell(\beta) = \sum_{i=1}^n (y_i x_i^T \beta - \log(1 + \exp(x_i^T \beta))). \quad (8)$$

Note that, because the prior is one, $f_{y, \beta}(y, \beta)$ equals the likelihood $f_{y|\beta}(y)$. We can thus calculate the ELBO as:

$$\begin{aligned} \text{ELBO}(q) &= \int q(\beta) \log \frac{f_{y, \beta}(y, \beta)}{q(\beta)} d\beta \\ &= \int q(\beta) \ell(\beta) d\beta - \int q(\beta) \log q(\beta) d\beta. \end{aligned}$$

The quantity $-\int q \log q$ is known as the entropy $H(q)$ of q . We thus have

$$\text{ELBO}(q) = \int q(\beta) \ell(\beta) d\beta - H(q).$$

Let us now take $\mathcal{Q}$ to be the class of all normal distributions $N(m, \Sigma)$ as $m \in \mathbb{R}^p$ and Σ varies over the class of all $p \times p$ positive definite matrices Σ . The entropy of the multivariate normal distribution (see https://en.wikipedia.org/wiki/Multivariate_normal_distribution) is:

$$H(N(m, \Sigma)) = \frac{p}{2} (1 + \log(2\pi)) + \frac{1}{2} \log |\Sigma|$$

where $|\Sigma|$ is the determinant of Σ . We thus have:

$$\text{ELBO}(m, \Sigma) = \mathbb{E}_{\beta \sim N(m, \Sigma)} \ell(\beta) + \frac{1}{2} \log |\Sigma| + \frac{p}{2} (1 + \log(2\pi)).$$

This can be maximized over m and Σ to obtain the posterior approximation $N(\hat{m}, \hat{\Sigma})$. Here m can be any vector over $\mathbb{R}^p$ but Σ is constrained to be positive definite. Constrained optimization is usually difficult (for gradient-based methods) so we convert this to unconstrained optimization by taking

$$\Sigma = LL^T$$

where L is a $p \times p$ lower-triangular matrix with non-zero diagonal entries. We can also, without loss of generality, assume that L has strictly positive diagonal entries (given any L ,

we can replace it by LD where D is a diagonal matrix with entries $+1$ if the corresponding entry of L is positive and -1 otherwise). In code, we can represent the diagonal entries of L by $\exp(a_1), \dots, \exp(a_p)$. With this parametrization, we have

$$|\Sigma| = |LL^T| = |L|^2 = \prod_{j=1}^p L_{jj}^2,$$

and thus

$$\text{ELBO}(m, L) = \mathbb{E}_{\beta \sim N(m, LL^T)} \ell(\beta) + \sum_{j=1}^p \log L_{jj} + \frac{p}{2} (1 + \log(2\pi)).$$

In order to maximize the above function, we need to make the term $\mathbb{E}_{\beta \sim N(m, LL^T)} \ell(\beta)$ more explicit. For this, we use the *reparametrization* trick (see https://en.wikipedia.org/wiki/Reparameterization_trick) and write

$$\beta = m + Lz \quad \text{where } z \sim N(0, I_p).$$

This gives

$$\text{ELBO}(m, L) = \mathbb{E}_{z \sim N(0, I_p)} \ell(m + Lz) + \sum_{j=1}^p \log L_{jj} + \frac{p}{2} (1 + \log(2\pi)).$$

The expectation above can be approximated by Monte Carlo. Specifically generate

$$z^{(1)}, \dots, z^{(S)} \stackrel{\text{i.i.d.}}{\sim} N(0, I_p)$$

and use the approximation:

$$\text{ELBO}(m, L) \approx \frac{1}{S} \sum_{s=1}^S \ell(m + Lz^{(s)}) + \sum_{j=1}^p \log L_{jj} + \frac{p}{2} (1 + \log(2\pi)).$$

This can be maximized using standard gradient-based optimization software such as PyTorch.

As an example of variational inference, we shall look at an example of Variational Auto-Encoders (VAE) which are used for generative modeling.

3 Variational AutoEncoders

Consider data points $y_1, \dots, y_n$, where each y_i is a binary vector in $\{0, 1\}^{d_y}$. One can think of each y_i as an image, flattened from its 2D pixel grid into a single vector of length d_y , where each entry corresponds to one pixel. In general, pixel values can take many forms—for instance, grayscale images use intensities in $\{0, 1, \dots, 255\}$, and color images use three such values per pixel (one per RGB channel). Here we focus on the simplest case: *binary* images (also called black-and-white or bilevel images), in which each pixel is either off (0) or on (1). For example, a 28×28 binary image yields $d_y = 28^2 = 784$.

We build up a model for the data in stages. Since each coordinate of y_i is binary, the most basic assumption is

$$y_i \mid p_i \stackrel{\text{independent}}{\sim} \text{Bernoulli}(p_i),$$

where $p_i \in [0, 1]^{d_y}$ specifies the probability that each pixel is on (Bernoulli is interpreted componentwise, so pixels are conditionally independent given p_i). In other words, we are assuming that each $y_i \in \{0, 1\}^{d_y}$ is Bernoulli with its own parameter $p_i \in [0, 1]^{d_y}$. We need to further model $p_1, \dots, p_n$ (otherwise, there will be too many parameters to yield anything useful). Here it is more convenient to work on the logit scale: let

$$\eta_i = \log \frac{p_i}{1 - p_i} \in \mathbb{R}^{d_y},$$

applied componentwise.

We shall assume $\eta_1, \dots, \eta_n$ are i.i.d with some common distribution on $\mathbb{R}^{d_y}$. Further, we assume that this common distribution, despite living in $\mathbb{R}^{d_y}$, is in fact concentrated on (or near) a much lower-dimensional set. A natural way to encode this is to write

$$\eta_i = f_\theta(z_i) \quad \text{where } z_i \stackrel{\text{i.i.d.}}{\sim} N(0, I_d),$$

where $f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^{d_y}$ is a neural network with parameters θ and $d \ll d_y$. The pushforward of $N(0, I_d)$ under f_θ defines a d -dimensional family of distributions on $\mathbb{R}^{d_y}$, and the flexibility of f_θ ensures this family is rich enough to model the structure we care about. The Gaussian prior on z_i is essentially a convention: any continuous distribution on $\mathbb{R}^d$ can be written as a deterministic transformation of a standard Gaussian, so fixing $z_i \sim N(0, I_d)$ and letting f_θ be flexible loses no generality (it is also analytically convenient, as we will see when designing an inference procedure). Putting the pieces together, and writing σ for the componentwise sigmoid, the overall model can be written in the following alternative way:

$$z_i \stackrel{\text{i.i.d.}}{\sim} N(0, I_d) \quad \text{and} \quad y_i \mid z_i \sim \text{Bernoulli}(\sigma(f_\theta(z_i)))$$

with, for example,

$$f_\theta(z_i) = B \text{ReLU}(Az_i + a) + b \quad \text{with } \theta = (A, a, B, b) \quad (9)$$

The unknown parameters in this model are the neural network parameters θ . The latent variables $z_1, \dots, z_n$ are also unobserved. The z_i 's can be integrated out and the model becomes

$$y_i \stackrel{\text{i.i.d.}}{\sim} p_\theta$$

where

$$p_\theta(y) = \int p_\theta(y \mid z) \varphi(z) dz$$

with $\varphi(z)$ being the density of $N(0, I_d)$, and

$$p_\theta(y \mid z) = \prod_{j=1}^{d_y} (\sigma(f_{\theta,j}(z)))^{y_j} (1 - \sigma(f_{\theta,j}(z)))^{1-y_j} \quad \text{for } y = (y_1, \dots, y_{d_y}) \in \{0, 1\}^{d_y}.$$

The likelihood then is given by:

$$f_{\text{data}|\theta}(y_1, \dots, y_n) = \prod_{i=1}^n \left[\int p_\theta(y_i \mid z_i) \varphi(z_i) dz_i \right]$$

and the corresponding log-likelihood is

$$\log f_{\text{data}|\theta}(y_1, \dots, y_n) = \sum_{i=1}^n \log \left[\int p_\theta(y_i \mid z_i) \varphi(z_i) dz_i \right]. \quad (10)$$

This log-likelihood cannot be directly used as the objective function in optimization software such as PyTorch because of the presence of the integral. A natural idea is to discretize the integral by sampling $z_i^{(l)}, l = 1, \dots, M$ from $N(0, I_d)$ and forming the approximate log-likelihood:

$$\log f_{\text{data}|\theta}(y_1, \dots, y_n) \approx \sum_{i=1}^n \log \left[\frac{1}{M} \sum_{l=1}^M p_{\theta}(y_i | z_i^{(l)}) \right]. \quad (11)$$

The above term (11) is unlikely to be a good approximation to the actual log-likelihood (10). This is because the integral $\int p_{\theta}(y_i | z_i) \varphi(z_i) dz_i$ in (10) is dominated by the region where $p_{\theta}(y_i | z_i)$ is large i.e., by the posterior $p_{\theta}(z_i | y_i)$. This posterior typically is very concentrated in the sense that only a tiny region of $z_i \in \mathbb{R}^d$ yields non-negligible values of $p_{\theta}(y_i | z_i)$. The prior $z_i \sim N(0, I_d)$, by contrast, spreads mass diffusely over $\mathbb{R}^d$. The consequence will be that almost every sample $z_i^{(l)}$ lands in a region where $p_{\theta}(y_i | z_i^{(l)})$ is small, so that the average in (11) is dominated by one or two (or even zero) lucky samples. This makes the average in (11) a highly variance estimate of the integral in (10), and hence unreliable as an optimization objective.

A better approach is to use ideas from variational inference and to maximize the ELBO instead.

4 Use of the ELBO for maximization of $\log f_{\text{data}|\theta}(y_1, \dots, y_n)$

We shall use the following formula:

$$\log f_{\text{data}|\theta}(y_1, \dots, y_n) = \max_q \text{ELBO}(q, \theta) \quad (12)$$

where

$$\begin{aligned} & \text{ELBO}(q, \theta) \\ &= \int \cdots \int q(z_1, \dots, z_n) \log \frac{\prod_{i=1}^n p_{\theta}(y_i, z_i)}{q(z_1, \dots, z_n)} dz_1 \dots dz_n \\ &= \int \cdots \int q(z_1, \dots, z_n) \log \prod_{i=1}^n p_{\theta}(y_i | z_i) dz_1 \dots dz_n - \int \cdots \int q(z_1, \dots, z_n) \log \frac{q(z_1, \dots, z_n)}{\varphi(z_1) \dots \varphi(z_n)} dz_1 \dots dz_n \\ &= \sum_{i=1}^n \int \log p_{\theta}(y_i | z_i) q_i(z_i) dz_i - KL(q \| \varphi \times \cdots \times \varphi). \end{aligned}$$

where q_i is the i -th marginal corresponding to q , and $\varphi \times \cdots \times \varphi$ denotes the distribution of $(z_1, \dots, z_n)$ corresponding to $z_i \stackrel{\text{i.i.d}}{\sim} N(0, I_d)$.

The maximization in (12) is over all probability densities q of $z_1, \dots, z_n$. We have also seen that the maximum in (12) is achieved when q is the conditional density of $z_1, \dots, z_n$ given $y_1, \dots, y_n$ and θ :

$$q^*(z_1, \dots, z_n) \propto \prod_{i=1}^n [p_{\theta}(y_i | z_i) \varphi(z_i)].$$

It is clear that this q^* is a product measure $\prod_{i=1}^n q_i^*(z_i | y_i)$ where

$$q_i^*(z_i | y_i) \propto p_{\theta}(y_i | z_i) \varphi(z_i) \implies q_i^*(z_i | y_i) = \frac{p_{\theta}(y_i | z_i) \varphi(z_i)}{\int p_{\theta}(y_i | z_i) \varphi(z_i) dz_i}.$$

However because of the somewhat complicated form of $p_\theta(y_i | z_i)$, the density q_i^* above is not of a simple form (for example, it is not a Gaussian). We therefore take a simpler class of densities $\mathcal{Q}$ and then maximize the ELBO over $\mathcal{Q}$. Specifically, we shall take $\mathcal{Q}$ to be the class of all product densities $\prod q_{i,\phi}(z_i | y_i)$ for which each marginal $q_\phi(z_i | y_i)$ is Gaussian:

$$q_\phi(z_i | y_i) = \text{density of } N(\mu_\phi(y_i), \Sigma_\phi(y_i))$$

with

$$\mu_\phi(y_i) = (\mu_{\phi,j}(y_i), j = 1, \dots, d) \quad \text{and} \quad \Sigma_\phi(y_i) = \text{diag}(\sigma_{\phi,j}^2(y_i), j = 1, \dots, d).$$

With this choice $\mathcal{Q}$, the ELBO becomes

$$\text{ELBO}(q, \theta) = \sum_{i=1}^n \int \log p_\theta(y_i | z_i) q_\phi(z_i | y_i) dz_i - \sum_{i=1}^n KL(q(\cdot | y_i) \| \phi),$$

where we use the fact that the KL between two product distributions becomes the sum of the KL between the marginals. We can further write this as:

$$\text{ELBO}(q, \theta) = \sum_{i=1}^n \mathbb{E}_{z_i \sim q_\phi(\cdot | y_i)} \log p_\theta(y_i | z_i) - \sum_{i=1}^n KL(N(\mu_\phi(y_i), \Sigma_\phi(y_i)) \| N(0, I_d)).$$

The second term above can be written in closed form as (see e.g., https://en.wikipedia.org/wiki/Kullback%E2%80%93Leibler_divergence#Examples)

$$KL(N(\mu_\phi(y_i), \Sigma_\phi(y_i)) \| N(0, I_d)) = \frac{1}{2} \sum_{j=1}^d \{ \sigma_{\phi,j}^2(y_i) + \mu_{\phi,j}^2(y_i) - \log \sigma_{\phi,j}^2(y_i) - 1 \}.$$

For the first term in the ELBO expression above, we need to compute

$$\mathbb{E}_{z_i \sim q_\phi(\cdot | y_i)} \log p_\theta(y_i | z_i)$$

For this, we can use Monte Carlo averaging. First write $z_i \sim N(\mu_\phi(y_i), \Sigma_\phi(y_i))$ as

$$z_i = \mu_\phi(y_i) + \sigma_\phi(y_i) \odot u_i \quad \text{where } u_i \stackrel{\text{i.i.d.}}{\sim} N(0, I_d).$$

Here $\odot$ stands for the Hadamard product (elementwise multiplication). Note that the components of $\Sigma_\phi(y_i)^{1/2} u_i$ are simply $\sigma_{\phi,j}(y_i) u_{i,j}$. Thus

$$\begin{aligned} \mathbb{E}_{z_i \sim q_\phi(\cdot | y_i)} \log p_\theta(y_i | z_i) &= \mathbb{E}_{u_i \sim N(0, I_d)} \log p_\theta(y_i | \mu_\phi(y_i) + \sigma_\phi(y_i) \odot u_i) \\ &\approx \frac{1}{L} \sum_{l=1}^L \log p_\theta(y_i | \mu_\phi(y_i) + \sigma_\phi(y_i) \odot u_i^{(l)}) \end{aligned}$$

where $u_i^{(l)}, 1 \leq l \leq L, 1 \leq i \leq n$ are i.i.d samples generated from $N(0, I_d)$. Thus the ELBO becomes:

$$\begin{aligned} \text{ELBO}(\phi, \theta) &= \frac{1}{L} \sum_{i=1}^n \sum_{l=1}^L \sum_{j=1}^{d_y} \left\{ y_{i,j} \log \sigma \left(f_{\theta,j}(\mu_\phi(y_i) + \sigma_\phi(y_i) \odot u_i^{(l)}) \right) \right. \\ &\quad \left. + (1 - y_{i,j}) \log \left[1 - \sigma \left(f_{\theta,j}(\mu_\phi(y_i) + \sigma_\phi(y_i) \odot u_i^{(l)}) \right) \right] \right\} \\ &\quad - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^d \{ \sigma_{\phi,j}^2(y_i) + \mu_{\phi,j}^2(y_i) - \log \sigma_{\phi,j}^2(y_i) - 1 \}. \end{aligned}$$

Here σ denotes the sigmoid function while σ_ϕ denotes the variance parameter of the variational distribution q_ϕ .

We simply maximize $\text{ELBO}(\phi, \theta)$ jointly over ϕ and θ to estimate them. We will parametrize μ_ϕ and σ_ϕ using neural networks. For example,

$$\begin{aligned}\mu_\phi(y) &= W_\mu \text{ReLU}(Wy + w) + b_\mu \\ \log \sigma_\phi^2(y) &= W_\sigma \text{ReLU}(Wy + w) + b_\sigma,\end{aligned}$$

with $\phi = (W_\mu, W, w, b_\mu, W_\sigma, b_\sigma)$. Recall also that f_θ will also be parametrized by a neural network as in (9).

With these parametrizations, $\text{ELBO}(\phi, \theta)$ can be maximized using optimization software such as PyTorch.